W9 Tensor Products

LXJ

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Chapter 2: Tensors and Differential Forms

The tensor product

Let U, V, W be finite dimensional real vector spaces, V^* the dual space of V.

Definition (Tensor product)

Let $F(V, W)$ be the free vector space over $\mathbb{R}$ generated by $V \times W$, and let $R(V, W)$ be the subspace generated by the elements

$$\begin{gathered} (v_1 + v_2, w) - (v_1, w) - (v_2, w) \\ (v, w_1 + w_2) - (v, w_1) - (v, w_2) \\ (av, w) - a(v, w), \qquad (v, aw) - a(v, w) \end{gathered} \quad \begin{pmatrix} a \in \mathbb{R} \\ v, v_1, v_2 \in V \\ w, w_1, w_2 \in W \end{pmatrix}.$$

The quotient space $F(V, W)/R(V, W)$ is called the *tensor product* of V and W, denoted $V \otimes W$. The coset of (v, w) is denoted $v \otimes w$.

Identities in $V \otimes W$

Thus we have the following identities in $V \otimes W$:

$$(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w,$$

$$v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2,$$

$$(av) \otimes w = a(v \otimes w), \qquad v \otimes (aw) = a(v \otimes w).$$

Note

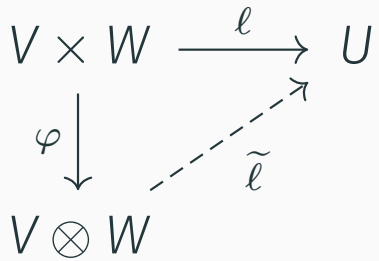
The tensor product is *bilinear* by construction: $\otimes : V \times W \to V \otimes W$ is linear in each argument separately.

Property 1: the universal mapping property

Let φ be the bilinear map

$$\varphi : V \times W \longrightarrow V \otimes W, \qquad (v, w) \longmapsto v \otimes w.$$

Then for any vector space U and any bilinear map $\ell : V \times W \to U$, there exists a *unique* linear map $\widetilde{\ell} : V \otimes W \to U$ s.t. the following diagram commutes:



The pair $(V \otimes W, \varphi)$ is said to solve the *universal mapping problem* for bilinear maps ℓ with domain $V \times W$.

Uniqueness and further properties

Moreover, $(V \otimes W, \varphi)$ are unique in the sense that if $(X, \widetilde{\varphi})$, $\widetilde{\varphi} : V \times W \to X$ a bilinear map, also solves the universal mapping problem, then there is an isomorphism $\alpha : V \otimes W \to X$ s.t. $\alpha \circ \varphi = \widetilde{\varphi}$.

(b) $V \otimes W$ is canonically isomorphic to $W \otimes V$ (both solve the universal mapping property);

(c) $V \otimes (W \otimes U)$ is canonically isomorphic to $(V \otimes W) \otimes U$.